

The Impact of Multipath Information on Time-of-Arrival Estimation

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Abstract—Time-of-arrival (TOA) based localization plays a central role in current and future localization systems. Such systems, exploiting the fine delay resolution properties of wideband and ultra-wideband (UWB) signals, are particularly attractive for ranging under harsh propagation conditions in which significant multipath may be present. While multipath has been traditionally considered detrimental in the design of TOA estimators, it can be exploited to benefit ranging. This paper investigates the impact of a priori multipath information on TOA estimation. To this end, bounds on the performance of TOA estimation in dense multipath channels are provided and discussed under different operating conditions. The effects of channel dispersion and transmission bandwidth on ranging systems are investigated showing that, contrary to communication systems, diversity behavior is only exhibited in the medium signal-to-noise ratio regime.

Index Terms—Ranging, time-of-arrival estimation, Ziv-Zakai bound, Cramér-Rao bound, ultra-wideband, localization

I. INTRODUCTION

LOCATION-AWARENESS is becoming increasingly important [1]–[12] in many commercial, public safety, and military applications, such as location-based services [13]–[16], and Internet-of-Things [17]–[24]. The coming years will see the emergence of applications which require localization systems to operate with sub-meter accuracy in harsh, cluttered propagation environments where Global Positioning System (GPS) typically fails [25]. For more information on ranging and localization and related methods we refer the reader to [26]–[38].

Among the different localization techniques, those based on time-of-arrival (TOA) measurements are particularly suitable for high localization accuracy [39]–[41]. In general, the ranging accuracy of TOA measurements increases with signal bandwidth and hence the use of wideband or ultra-wideband

(UWB) signals [42]–[44] is attractive for ranging applications [45]–[51]. Therefore, UWB signaling offers the potential of achieving high accuracy localization even in cluttered environments [52]–[54], due to its ability to resolve closely-spaced multipath components (MPCs) [55]–[57].

These TOA estimation techniques typically operate by estimating the time at which the first signal path arrives, and are primarily affected by noise, multipath propagation, obstruction of the direct path, interference, and clock drift [49]. In realistic propagation environments the first multipath component is frequently not the strongest, making TOA estimation even more challenging and calling for techniques to cope with such impairments [58]–[60].

While the performance of a specific TOA estimation algorithm can be evaluated, estimation error bounds play a fundamental role since they can serve as useful performance benchmarks for the design of TOA estimators. As in communication systems the channel condition drives the ultimate performance, the same is true also in TOA estimation. When the relationship between the MPCs and the position of transmitter and receiver is not known (either exactly or statistically), the multipath does not contribute any further useful information to the estimation process beyond what is available from the first (direct) path. Hence, the MPCs can be considered as nuisance parameters to be jointly estimated. On the contrary, when a priori information about the geometric or electromagnetic characteristics of the environment is available, the MPC characteristics (e.g., specular components or statistics) can be linked to the position/distance. As a consequence, it is reasonable to expect that some further information can be distilled from MPCs during the estimation process. One potential approach would be to use multipath-assisted localization techniques [61]–[64], typically based on simultaneous localization and mapping. Therefore, to get full insight into the real fundamental limits in TOA estimation, the channel characteristics and any a priori information on MPC should be properly accounted for in the derivation of performance bounds.

In practice, a priori information (deterministic or statistical) on MPCs can be acquired in different ways. One possibility is to associate the multipath components to the local geometry using a known floor plan, if available, as done in [62]. If the floor plan is not available, crowd-based learning approaches, such as those presented in [20], could be adopted to learn how the channel MPCs depend on the position of the transmitter and receiver. Initially, no a priori information is available during the TOA estimation and hence a rough position estimate is obtained and the associated MPCs are learned. When new

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users enter the same area, they can take advantage of the learned MPCs as a priori information during TOA estimation, likely resulting in improved accuracy. This process is then repeated for subsequent users, providing continual refinement of the a priori information. Under a similar philosophy, simultaneous localization and mapping approaches could be exploited to further refine the TOA estimation in an iterative way once rough position information of the mobile user and the reference nodes is obtained.

In the end, regardless of the approach adopted, perfect deterministic or statistical information on MPCs cannot be obtained. The objective of this paper is to derive bounds that can incorporate the available a priori information and serve as benchmarks of any practical estimator.

A. The Cramér-Rao bound (CRB)

The CRB has been widely used as a performance benchmark for assessing estimation error. Its use is justified by asserting that the performance of the maximum likelihood (ML) estimator approaches the CRB asymptotically for sufficiently long observation times or for high signal-to-noise ratios (SNRs) [65]. It is well known, however, that the CRB is not accurate at low and moderate SNR. In fact, the performance of the TOA estimator, like all non-linear estimators, is characterized by the presence of distinct SNR regions (i.e., low, medium, and high SNRs) corresponding to different modes of operation. This behavior is referred to as the *threshold effect* and it has been studied in a variety of contexts (see, for example, [66]–[72]). In the low SNR region (also known as the *a priori region*), the mean square error (MSE) is close to that obtained solely from the a priori information about the TOA and signal observations provide little additional information. In the high SNR region (also known as the *asymptotic region*), the MSE is accurately predicted by the CRB. Between these two extremes, there may be an additional region (also known as the *transition region* or *ambiguity region*) where observations are subject to ambiguities that are not accounted for by the CRB [65], but could be highly useful in many contexts.

B. The Ziv-Zakai bound (ZZB)

Although the CRB accurately predicts the MSE in the asymptotic region, operating in such a region is often impractical due to constraints on observation time or power levels. Therefore, other bounds, which are more complicated but tighter than the CRB, have been proposed in the literature. In particular, the Barankin bound identifies the SNR values (thresholds) that distinguish the ambiguity region [71], [72]. The Ziv-Zakai bound (ZZB) [73], with its improved versions such as the Bellini-Tartara Ziv-Zakai bound (BTZZB) [74], [75] and the Chazan-Zakai-Ziv bound [76], as well as the Weiss-Weinstein bound (WWB) [69] are more accurate alternatives to the Barankin bound. They can be applied to a wider range of SNRs and account for both ambiguity effects and a priori information on the parameter to be estimated. However, in many cases they may not be analytically tractable [77] or require more complicated evaluations compared to the CRB,

especially when operating in the presence of multipath [78]–[82]. The ZZB has been applied in several contexts, including: symbol timing recovery [83], [84], direct position estimation [85], and in visible light positioning systems [86].

C. CRBs and ZZBs for TOA Estimation with Multipath

Several other works have considered bounds for various applications in multipath channels under different assumptions for type of channels, knowledge of the channel, and presence of interference. In [30], [87], the expression of the CRB for multipath environments is given starting from the joint estimation of channel parameters considered as unknown and deterministic. It is shown that the performance is only affected by the first cluster of MPCs overlapping the first path and hence no additional information is carried by the remaining MPCs. The CRB for multi-carrier signals in the presence of interference is presented in [88]. A survey on the CRB for parameter estimation under misspecified models can be found in [89].

Aspects of ZZB have been investigated in a variety of scenarios in the literature. The classical papers on ZZB for the estimation of the relative delay between two sensors observing a source emitting noise-like signals [66], [67] are extended to UWB signals in [81]. The ZZB for Gaussian signals assuming perfect channel knowledge at the receiver is evaluated in [90]. The concept of the modified Cramér-Rao bound (MCRB), introduced in [91], [92] to the evaluation of the bound averaged over the nuisance parameters, is used in [93] to derive the modified ZZB and it is applied to satellite positioning systems. In [82], the ZZB accounting for the geometry of an indoor environment is derived for the UWB ranging error under IEEE 802.15.4a channel models. The bound can also be evaluated using the channel measurements from real environments. In [94] the ZZB in the presence of unknown interference is derived.

A few results are presented for the case where the receiver has partial or no knowledge about the channel [78]–[80]. In particular, in [78] the BTZZB is investigated using measured data as well as Monte Carlo generated channel pulse responses (CPRs). In [79], the ZZB is derived using a second-order statistics approach by modeling the received signal as a non-stationary Gaussian random process. A ZZB that does not assume the knowledge of the channel realization and incorporates a priori information for the random Gaussian channel is developed in [80]. The expressions derived for the ZZB are complex and require the evaluation of a three-fold integral which causes some numerical computation issues.

D. Our Contribution

This paper investigates TOA estimation error bounds for wideband and UWB ranging systems under different a priori information on the multipath phenomena. The main contributions of this paper are as follows:

- We derive analytical expressions for the CRB and ZZB on the root mean square error (RMSE) of TOA estimation under different a priori multipath information;

- We demonstrate that the ZZB reveals insights on the information carried by the multipath that is not captured by the classical CRB and provides improved bounds for a wider range of SNRs; and
- We investigate the impact of the a priori multipath information on TOA estimation, characterizing the intertwined effect of diversity and CPR dispersion on ranging systems.

As will be clarified later, as a byproduct of this paper, the optimal binary pulse position modulation (PPM) Rake receiver is designed.

The remainder of this paper is organized as follows. In Sec. II the signal and channel models are introduced. The ZZB is reviewed briefly in Sec. III. The CRB and ZZB are derived for realistic multipath environments under different operating conditions in Secs. IV and V. Numerical results and discussions are presented in Sec. VI. Finally, conclusions are given in Sec. VII.

Throughout we use the following notation: random variables (RVs) and processes are displayed in sans serif, upright fonts; their realizations in serif, italic fonts. Vectors and matrices are denoted by bold lowercase and uppercase letters, respectively. For example, a RV and its realization are denoted by $\mathbf{x}$ and x ; a random vector and its realization are denoted by $\mathbf{x}$ and $\mathbf{x}$. The functions $f_{\mathbf{x}}(\mathbf{x}|\theta)$ and $f_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y};\theta)$ denote the probability density function (PDF) of $\mathbf{x}$ and conditional PDF of $\mathbf{x}$ given $\mathbf{y}$, respectively, with θ denoting a parameter of the distribution. When the role of parameter θ is not relevant, the conditional PDF will be denoted simply by $f_{\mathbf{x}|\mathbf{y}}(\mathbf{x}|\mathbf{y})$. Expectation of the random variable, random vector, or random matrix in the argument is denoted by $\mathbb{E}\{\cdot\}$. Variance of the random variable in the argument is denoted by $\mathbb{V}\{\cdot\}$. Probability of the event in the argument is denoted by $\mathbb{P}\{\cdot\}$. The real and imaginary parts of x are denoted by $\Re\{x\}$ and $\Im\{x\}$, respectively. Finally, the operator $\otimes$ represents the Kronecker product.

II. SIGNAL AND CHANNEL MODELS

We consider the transmission of a single band-limited pulse or waveform $p(t)$ with bandwidth W and energy E_p .¹ In practice, $p(t)$ is generated starting from a baseband pulse $g(t)$ that is modulated to a center frequency f_c :

$$p(t) = \sqrt{2E_p} \Re\{g(t)e^{j2\pi f_c t}\} \quad (1)$$

where $g(t)$ has unit energy. We define

$$\rho_p(z) = \frac{1}{E_p} \int_{-\infty}^{\infty} p(t) p^*(t-z) dt \quad (2)$$

as the normalized autocorrelation function of $p(t)$,² and

$$\beta_p = \sqrt{\frac{\int_{-\infty}^{\infty} f^2 |P(f)|^2 df}{E_p}} \quad (3)$$

as the effective bandwidth of $p(t)$, where $P(f)$ is the Fourier transform of $p(t)$. The normalized autocorrelation function

$\rho_g(z)$ and the effective bandwidth β_g of $g(t)$ can be defined in a similar way. We consider the case when $g(t)$ is a real signal and therefore we can write

$$p(t) = \sqrt{2E_p} g(t) \cos(2\pi f_c t). \quad (4)$$

Note that when $g(t)$ is an even function, as happens in most of practical cases, we have $\beta_p^2 = f_c^2 + \beta_g^2$. When $p(t)$ is transmitted through a multipath channel, the received signal can be represented as

$$r(t) = s(t - \tau) + n(t) \quad (5)$$

where $s(t)$ is the CPR, τ is the TOA of the received signal, and $n(t)$ is zero-mean additive white Gaussian noise (AWGN) with two-sided spectral density $N_0/2$.

The goal is to obtain an estimate $\hat{\tau}$ of τ by observing $r(t)$ in the interval $[0, T_{ob})$ in the presence of signal distortion due to multipath. In the absence of other information, we assume τ to be unknown with domain³ given by the interval $[0, T_a)$, with $T_a + T_d < T_{ob}$, where T_d is the dispersion of the channel, so that all the multipath components fall within the observation interval T_{ob} .

Several techniques can be adopted to solve this problem (see for example [49] and the references therein). In this paper, we derive and discuss the CRB and improved bounds based on the ZZB for TOA estimation in dense multipath channels under different a priori information.

We consider a conventional multipath channel model with M physical paths given by

$$h(t) = \sum_{m=1}^M \alpha_m \delta(t - \tau_m) \quad (6)$$

where $\delta(t)$ is the Dirac delta pseudo-function, α_m and τ_m are, respectively, the real gain and the relative delay of the m th multipath component. Delays $\{\tau_m\}$ are relative to the TOA τ of the first multipath component, and hence we set $\tau_1 = 0$. After the transmission of the bandpass pulse, $p(t)$, the response $s(t)$ can be expressed as the output of a tapped-delay line equivalent channel [95]–[99] with sampling interval $\Delta\tau \triangleq 1/W$

$$s(t) = \sqrt{2E_p} \sum_{l=1}^L [\nu_l \cos(2\pi f_c t) - \nu_{L+l} \sin(2\pi f_c t)] \times g(t - (l-1)\Delta\tau) \quad (7)$$

where L is chosen to capture the multipath spread of the channel, $\nu_l = h_I(l\Delta\tau)$, and $\nu_{L+l} = h_Q(l\Delta\tau)$. The quantities $h_I(\cdot)$ and $h_Q(\cdot)$ are defined as the in-phase and quadrature components, respectively, of the baseband channel observed with a bandwidth of $W/2$, and are given by

$$h_I(t) = \sum_m \alpha_m \cos(2\pi f_c \tau_m) \text{sinc}(W(t - \tau_m)) \quad (8)$$

$$h_Q(t) = \sum_m \alpha_m \sin(2\pi f_c \tau_m) \text{sinc}(W(t - \tau_m)). \quad (9)$$

In a dense multipath scenario, (8) and (9) indicate that the samples of $h_I(\cdot)$ and $h_Q(\cdot)$ contain contributions from many

¹In general $p(t)$ can represent a multiple access signal such as direct sequence or time-hopping (TH).

²Since a band-limited signal cannot also be time-limited, we assume it can be truncated in the interval $[0, T_p]$ with some approximation error.

³Such a priori information could be the result of an initial coarse search.

paths. Under an uncorrelated scattering assumption, each of the samples will consist of many independent contributions. Hence, we model the samples of $h_l(\cdot)$ and $h_Q(\cdot)$ as independent Gaussian random variables [100]–[102].

For simplicity we will continue to use the word “path” instead of tap even though it may be the result of the contribution of multiple physical paths according to (8) and (9). We denote the average power gain of the l th path as $\Gamma_l = \mathbb{E}\{v_l^2\} + \mathbb{E}\{v_{L+l}^2\}$, where v_l and v_{L+l} are defined after (7). Without loss of generality we assume $\sum_{l=1}^L \Gamma_l = 1$, so that E_p also represents the average received energy per transmitted pulse. For each set of channel parameters $\mathbf{v} = [v_1, v_2, \dots, v_{2L}]^T$ we obtain a realization $s(t, \mathbf{v})$ of the random process characterizing the received signal. We denote the joint PDF of the vector $\mathbf{v}$ as $f_{\mathbf{v}}(\mathbf{v})$.

The above considerations lead to the widely used Ricean/Rayleigh multipath channel model, where $f_{\mathbf{v}}(\mathbf{v})$ is a $2L$ -dimensional joint Gaussian PDF. The v_l are independent RVs with

$$\mu_l = \mathbb{E}\{v_l\} = \begin{cases} s_l & 1 \leq l \leq L \\ 0 & L < l \leq 2L \end{cases} \quad (10)$$

and variance $\mathbb{V}\{v_l\} = \mathbb{V}\{v_{L+l}\} = \sigma_l^2$, $1 \leq l \leq L$. The parameters s_l and $2\sigma_l^2$ are the average amplitude of the specular component (when present) and the power gain of the scattered component of the l th path, respectively. Thus, the average power gain of the l th path is $\Gamma_l = \mu_l^2 + 2\sigma_l^2$. For Ricean fading, we define the Ricean factor or the power ratio between the specular and the scattered signals for individual paths as $K_l = s_l^2/(2\sigma_l^2)$, $1 \leq l \leq L$. For notational convenience we define the following vectors and matrices of the channel parameters: $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_{2L}]^T$, $\boldsymbol{\Psi} = \text{diag}\{\sigma_1^2, \sigma_2^2, \dots, \sigma_L^2\}$, and $\boldsymbol{\Psi}_2 = \mathbf{I}_2 \otimes \boldsymbol{\Psi}$, where $\mathbf{I}_n$ is the $n \times n$ identity matrix.

III. THE ZIV-ZAKAI LOWER BOUND

We briefly review the scalar ZZB to facilitate the mathematical development in the subsequent sections.⁴ The Ziv-Zakai lower bound can be derived starting from the following general identity for estimation MSE⁵

$$\text{MSE} = \mathbb{E}\{\varepsilon^2\} = \frac{1}{2} \int_0^\infty z \mathbb{P}\left\{|\varepsilon| \geq \frac{z}{2}\right\} dz \quad (11)$$

where $\varepsilon \triangleq \hat{\tau} - \tau$ represents the estimation error. A lower bound on the MSE can then be obtained by lower-bounding the quantity $\mathbb{P}\{|\varepsilon| \geq z/2\}$ [74], [75]. It can be shown that

$$\mathbb{P}\left\{|\varepsilon| \geq \frac{z}{2}\right\} = \int_{-\infty}^\infty \left[f_{\tau}(\zeta) \mathbb{P}\left\{\hat{\tau} \geq \zeta + \frac{z}{2} \mid \tau = \zeta\right\} + f_{\tau}(\zeta + z) \mathbb{P}\left\{\hat{\tau} \leq \zeta + \frac{z}{2} \mid \tau = \zeta + z\right\} \right] d\zeta \quad (12)$$

⁴Specifically, we consider the improved version of the ZZB described by Bellini-Tartara [74], [75]. A more appropriate acronym would be BTZZB, however we use ZZB for conciseness.

⁵Here the expectation is with respect to τ and $r(t)$.

where $f_{\tau}(\cdot)$ is the PDF of the TOA τ . The previous expression can be lower-bounded by

$$\int_{-\infty}^\infty 2 \min\left\{f_{\tau}(\zeta), f_{\tau}(\zeta + z)\right\} \times \left[\frac{1}{2} \mathbb{P}\left\{\hat{\tau} \geq \zeta + \frac{z}{2} \mid \tau = \zeta\right\} + \frac{1}{2} \mathbb{P}\left\{\hat{\tau} \leq \zeta + \frac{z}{2} \mid \tau = \zeta + z\right\} \right] d\zeta. \quad (13)$$

The term in square brackets in (13) represents the probability of error for deciding between two equally probable hypotheses, i.e.,

$$\begin{aligned} H_1 : \tau &= \zeta \\ H_2 : \tau &= \zeta + z \end{aligned} \quad (14)$$

according to a suboptimal rule. This rule requires that the parameter τ is first estimated and then the decision between the two hypothesis is made based on a minimum distance criterion as described in [76]. Thus, (13) can be further lower-bounded by replacing the term in square brackets with the error probability $P_{\min}(\zeta, \zeta + z)$ of the optimum decision rule that minimizes the error probability. The corresponding likelihood ratio test (LRT) is

$$\frac{f_{\mathbf{r}|\tau}(\mathbf{r}|\zeta)}{f_{\mathbf{r}|\tau}(\mathbf{r}|\zeta + z)} \underset{H_2}{\overset{H_1}{\gtrless}} 1 \quad (15)$$

where $\mathbf{r}$ is the vector representation of $r(t)$ in (5) obtained via Karhunen-Loève expansion [65]. In (15), $f_{\mathbf{r}|\tau}(\mathbf{r}|\tau)$ is considered a function of τ for a fixed $\mathbf{r}$ and is a likelihood function. Intuitively, this function quantifies the likelihood of τ given an observed waveform $r(t)$.

Under the assumption $T_a + T_d < T_{\text{ob}}$, $P_{\min}(\zeta, \zeta + z)$ does not depend on the delay ζ , thus it can be written as $P_{\min}(z)$ and the bound can be formulated in terms of z [76]. When τ is uniformly distributed in $[0, T_a)$, the ZZB is given by [74]–[76]

$$\eta_{\text{ZZ}} = \frac{1}{T_a} \int_0^{T_a} z (T_a - z) P_{\min}(z) dz. \quad (16)$$

The main challenge in evaluating (16) is to design the binary detection scheme based on the likelihood ratio in (15) and to derive a tractable expression for its performance $P_{\min}(z)$. Interestingly, we observe that the derivation of the ZZB has been transformed to the evaluation of the performance $P_{\min}(z)$ of a binary detection problem. We will exploit this observation in subsequent sections to obtain the ZZB for multipath environments.

IV. RECEIVER WITH PERFECT KNOWLEDGE OF THE CPR

Although it may be unrealistic to assume that the receiver has perfect knowledge of the CPR, this case can provide insights into the TOA estimation performance. In addition, it is not completely unrealistic since it has recently been shown in [61]–[63] that the specular components of the multipath can be predicted in practice given certain a priori knowledge of the environment or using simultaneous localization and mapping approaches. Any TOA estimator we design can never be better than a hypothetical estimator in which the receiver first measures the CPR perfectly, and then performs the estimation conditioned on that CPR. In this case, the bound

is derived by first conditioning on the channel realization (i.e., the nuisance parameters), and then averaging the result over all possible channel realizations. We refer to this approach as the *conditioning approach* [77] which results in the *perfect channel knowledge bound*.

A. Cramér-Rao Bound

For the derivation of the CRB, we consider $\tau = \tau$ to be unknown and deterministic. The first step is to obtain the expression for the CRB conditioned on the CPR $s(t)$, or equivalently its vector representation $\mathbf{s}$, as

$$\eta_{\text{CR}}|_{\mathbf{s}=\mathbf{s}} = \left[-\mathbb{E} \left\{ \frac{\partial^2}{\partial \tau^2} \ln f_{\mathbf{r}|\mathbf{s}}(\mathbf{r}|\mathbf{s}; \tau) \right\} \right]^{-1} \quad (17)$$

where $f_{\mathbf{r}|\mathbf{s}}(\mathbf{r}|\mathbf{s}; \tau)$ is given by [65]

$$f_{\mathbf{r}|\mathbf{s}}(\mathbf{r}|\mathbf{s}; \tau) \propto \exp \left\{ \int_0^{T_{\text{ob}}} \frac{2s(t-\tau)r(t) - s^2(t-\tau)}{N_0} dt \right\}. \quad (18)$$

Substituting (18) into (17), it can be shown that [103]

$$\eta_{\text{CR}}|_{\mathbf{s}=\mathbf{s}} = \frac{1}{8\pi^2 \beta_s^2 \gamma_{\text{snr}}} \quad (19)$$

where β_s is the effective bandwidth of $s(t)$ and $\gamma_{\text{snr}} \triangleq E_p/N_0$ is the average SNR. By substituting (7) in (18) and then evaluating (17), we can equivalently write

$$\eta_{\text{CR}}|_{\mathbf{v}=\mathbf{v}} = \frac{1}{2\mathbf{v}^T \boldsymbol{\Upsilon} \mathbf{v} \gamma_{\text{snr}}} \quad (20)$$

where

$$\boldsymbol{\Upsilon} = \begin{bmatrix} \dot{\mathbf{R}} + (2\pi f_c)^2 \mathbf{R}(0) & 4\pi f_c \mathbf{R}_{g\dot{g}} \\ -4\pi f_c \mathbf{R}_{g\dot{g}} & \dot{\mathbf{R}} + (2\pi f_c)^2 \mathbf{R}(0) \end{bmatrix}. \quad (21)$$

Here, $\mathbf{R}(z)$ is a matrix with elements $[\mathbf{R}(z)]_{kl} = \rho_g(z + (l-k)\Delta\tau)$, $\dot{\mathbf{R}}$ is a symmetric matrix with elements $[\dot{\mathbf{R}}]_{kl} = \int \dot{g}(t - (k-1)\Delta\tau)\dot{g}(t - (l-1)\Delta\tau) dt$, and $\mathbf{R}_{g\dot{g}}$ is a skew-symmetric matrix with elements $[\mathbf{R}_{g\dot{g}}]_{kl} = \int g(t - (k-1)\Delta\tau)\dot{g}(t - (l-1)\Delta\tau) dt$.

The next step is to average (20) over the nuisance parameters $\mathbf{v}$. A related approach was adopted in [104], where averaging of (19) was carried out using CPRs generated according to IEEE 802.15.3a UWB channel models in Monte Carlo iterations. Averaging (20) can give misleading results since the CRB is meaningful only for large values of the instantaneous SNR (i.e., asymptotic regime) given by the quantity $\mathbf{v}^T \boldsymbol{\Upsilon} \mathbf{v} \gamma_{\text{snr}}$ in (20). In other words, for certain realizations of the channel parameters $\mathbf{v}$, the denominator of (20) assumes low values resulting in high contributions during the averaging process. This can potentially lead to inconsistent results, that is, a bound larger than the initial unknown parameter uncertainty level T_a . Alternatively, the use of Jensen's inequality results in the MCRB [91] as follows

$$\begin{aligned} \mathbb{E}_{\mathbf{v}}\{\eta_{\text{CR}}|_{\mathbf{v}}\} &\geq \frac{1}{2\gamma_{\text{snr}} \mathbb{E}\{\mathbf{v}^T \boldsymbol{\Upsilon} \mathbf{v}\}} \\ &= \frac{1}{2\gamma_{\text{snr}} [(2\pi)^2 (\beta_g^2 + f_c^2) \text{tr}\{\boldsymbol{\Psi}\} + \boldsymbol{\mu}^T \boldsymbol{\Upsilon} \boldsymbol{\mu}]} \triangleq \eta_{\text{MCR}} \quad (22) \end{aligned}$$

where $\text{tr}\{\cdot\}$ is the trace operator and we have used the fact that $\mathbb{E}\{\mathbf{v}\mathbf{v}^T\} = \boldsymbol{\Psi}_2 + \boldsymbol{\mu}\boldsymbol{\mu}^T$ and $[\dot{\mathbf{R}}]_{kk} = (2\pi)^2 \beta_g^2$. Notice that the denominator of (22) increases with the SNR, effective bandwidth β_g , and center frequency f_c . Thus, a large effective bandwidth and/or large center frequency is beneficial for ranging, justifying the adoption of wideband or, better yet, UWB signals to achieve accurate ranging.

Note that the MCRB is still a bound and, in some cases, it reduces to the expression for AWGN, (i.e., no multipath or fading), [103]

$$\eta_{\text{CR}}^{(\text{AWGN})} = \frac{1}{8\pi^2 \beta_p^2 \gamma_{\text{snr}}}. \quad (23)$$

These cases are: 1) when there is no mean component ($\boldsymbol{\mu} = \mathbf{0}$), 2) when μ_l and μ_{L+l} are non-zero for only one $l \in \{1, 2, \dots, L\}$, or 3) when adjacent pulses are resolvable (resolvable multipath). Details are shown in Appendix I. As expected, under perfect channel knowledge, all MPCs can be exploited to improve the performance (please note that γ_{snr} in (23) is by definition the total SNR, and hence all the MPCs contribute). One may compare (23) with the CRB associated to the opposite scenario in which no a priori knowledge of the channel is available and all MPCs have to be jointly estimated. Assuming no overlap of the MPCs with the first path, we have [30]

$$\eta_{\text{CR}}^{(\text{p})} = \frac{1}{8\pi^2 \beta_p^2 \gamma_p} \quad (24)$$

where $\gamma_p = E_p \Gamma_1/N_0 = \gamma_{\text{snr}} \Gamma_1$ is the SNR related to the first path only. Here the MPCs do not contribute any additional information to the estimation, and thus the ratio $\gamma_{\text{snr}}/\gamma_p$ gives the gain obtained when perfect knowledge of the MPCs is available and exploited by the estimator.

Note that the bounds derived so far assume that $s(t)$, and hence the transmitted pulse $p(t)$, are perfectly known at the receiver. Even in AWGN, the CRB given by (23) is meaningful only when $p(t)$ is completely known at the receiver and the phase of the transmitted pulse has been retained at the receiver (coherent receiver). When the phase of the transmitted pulse is uncontrolled (noncoherent transmitter)⁶ or the estimator is not coherent, the phase of the transmitted pulse cannot be used to provide additional information about the TOA. Hence, estimation of the TOA must rely only on the signal envelope. In this case the CRB has to be evaluated considering the equivalent lowpass received signal instead of the bandpass signal. This leads to an expression similar to (23) with β_p replaced by β_g , the effective bandwidth of the baseband pulse $g(t)$ [103], and the resulting bound depends only on the signal bandwidth and not on the center frequency f_c . As will be seen in the numerical results, fading effects of the channel could also produce, in some configurations, uncorrelated random phase variations. In these cases, the performance is dependent only on the signal envelope indicating that coherent receivers provide little benefit.

⁶For example, random variations of the carrier phase will occur if the local oscillator used to generate $p(t)$ is not synchronized with the generator of the baseband pulse $g(t)$. See [103] for a discussion regarding the relationship between the CRB for the coherent and noncoherent cases.

B. Ziv-Zakai Bound

As discussed in Sec. III, the ZZB requires the evaluation of the error probability $P_{\min}(z)$ for deciding between the two hypothesis in (14). The test we design can never be better than a “genie test” in which a genie provides the receiver with the perfect CPR $s(t)$ (or, equivalently, $\mathbf{v}$) and then the receiver performs the LRT conditioned on that CPR. This is equivalent to evaluating the error probability $P_{\min}(z, \mathbf{v})$ corresponding to a classical binary communication system employing the real waveforms $s(t - \zeta, \mathbf{v})$ and $s(t - \zeta - z, \mathbf{v})$ for hypothesis H_1 and H_2 , respectively, with known $\mathbf{v}$. Therefore, the error probability is given by [74], [75]

$$P_{\min}(z, \mathbf{v}) = Q\left(\sqrt{\gamma_{\text{snr}}(\rho_s(0, \mathbf{v}) - \rho_s(z, \mathbf{v}))}\right) \quad (25)$$

where $Q(\cdot)$ is the Gaussian Q -function and we have defined the normalized autocorrelation function as

$$\begin{aligned} \rho_s(z, \mathbf{v}) &= \frac{1}{E_p} \int_{-\infty}^{\infty} s(t - \zeta, \mathbf{v}) s(t - \zeta - z, \mathbf{v}) dt \\ &= \mathbf{v}^T \mathbf{R}_2(z) \mathbf{v} \end{aligned} \quad (26)$$

and

$$\mathbf{R}_2(z) \triangleq \left(\begin{bmatrix} \cos(2\pi f_c z) & \sin(2\pi f_c z) \\ -\sin(2\pi f_c z) & \cos(2\pi f_c z) \end{bmatrix} \otimes \mathbf{R}(z) \right) \quad (27)$$

obtained by substituting (7) for a known $\mathbf{v}$ in (26). By replacing $P_{\min}(z)$ with $P_{\min}(z, \mathbf{v})$ in (16), we obtain the ZZB conditioned on $\mathbf{v}$ as

$$\eta_{\text{ZZ}}|_{\mathbf{v}=\mathbf{v}} = \frac{1}{T_a} \int_0^{T_a} z (T_a - z) P_{\min}(z, \mathbf{v}) dz. \quad (28)$$

Starting from (28), the unconditional ZZB can be evaluated by averaging over $\mathbf{v}$, that is,

$$\eta_{\text{ZZ}} = \int_{\mathbb{R}^{2L}} f_{\mathbf{v}}(\mathbf{v}) \eta_{\text{ZZ}}|_{\mathbf{v}=\mathbf{v}} d\mathbf{v} = \frac{1}{T_a} \int_0^{T_a} z (T_a - z) \bar{P}_{\min}(z) dz \quad (29)$$

where

$$\bar{P}_{\min}(z) \triangleq \int_{\mathbb{R}^{2L}} f_{\mathbf{v}}(\mathbf{v}) P_{\min}(z, \mathbf{v}) d\mathbf{v}. \quad (30)$$

We stress that (29) assumes that the receiver has perfect knowledge of the CPR, and hence serves as a lower bound on the MSE of the TOA estimation when the receiver measures the CPR perfectly. Note that (30) is the average bit error probability (BEP) for coherent detection of binary PPM employing a Rake receiver with perfect channel state information in the presence of multipath. The intertwined relationship between the BEP for PPM-based systems and the ZZB has been investigated in the literature under different conditions (e.g., multipath and interference) [90], [105], [106]. Similar to [90], we take an approach that leverages the well-known identity [107]:

$$Q(x) = \frac{1}{\pi} \int_0^{\pi/2} \exp\left\{-\frac{x^2}{2\sin^2\theta}\right\} d\theta. \quad (31)$$

Substituting (25) into (30) together with (31) gives

$$\begin{aligned} \bar{P}_{\min}(z) &= \mathbb{E}_{\mathbf{v}}\{P_{\min}(z, \mathbf{v})\} \\ &= \frac{1}{\pi} \int_0^{\pi/2} \mathbb{E}_{\mathbf{v}}\left\{\exp\left\{-\frac{\gamma_{\text{snr}} \mathbf{v}^T [\mathbf{R}_2(0) - \mathbf{R}_2(z)] \mathbf{v}}{2\sin^2\theta}\right\}\right\} d\theta \\ &= \frac{1}{\pi} \int_0^{\pi/2} \Phi_{\rho}\left(-\frac{\gamma_{\text{snr}}}{2\sin^2\theta}\right) d\theta \end{aligned} \quad (32)$$

where we have replaced the inner expectation in (32) with the evaluation of the moment generating function (MGF) of the scalar quantity $\rho \triangleq \mathbf{v}^T (\mathbf{R}_2(0) - \mathbf{R}_2(z)) \mathbf{v} = \mathbf{v}^T \mathbf{T}(z) \mathbf{v}$, defined as $\Phi_{\rho}(\xi) \triangleq \mathbb{E}\{\exp\{\xi\rho\}\}$. Here $\mathbf{T}(z)$ is a symmetric matrix given by $\mathbf{T}(z) = \mathbf{R}_2(0) - \frac{1}{2}(\mathbf{R}_2(z) + \mathbf{R}_2^T(z))$. Using the results of [108] the MGF of ρ can be derived as

$$\Phi_{\rho}(\xi) = \frac{\exp\left\{\xi \boldsymbol{\mu}^T \mathbf{T}(z) [\mathbf{I}_{2L} - 2\xi \boldsymbol{\Psi}_2 \mathbf{T}(z)]^{-1} \boldsymbol{\mu}\right\}}{\det\{\mathbf{I}_{2L} - 2\xi \boldsymbol{\Psi}_2 \mathbf{T}(z)\}^{1/2}}. \quad (33)$$

We note that the evaluation of $\bar{P}_{\min}(z)$ no longer requires the $2L$ -fold multidimensional integration of (30); instead, we need only to evaluate a single integral with finite limits. Efficient and accurate methods to evaluate integrals in the form of (32) can be found in [109].

In AWGN the ZZB for the perfect CPR case reduces to [76]

$$\eta_{\text{ZZ}}^{(\text{AWGN})} = \frac{1}{T_a} \int_0^{T_a} z (T_a - z) Q\left(\sqrt{\gamma_{\text{snr}}(1 - \rho_p(z))}\right) dz. \quad (34)$$

The ZZBs for noncoherent receivers are investigated in [110].

V. RECEIVER WITH STATISTICAL KNOWLEDGE OF THE CPR

When available, a priori statistical information on the CPR can be exploited in the estimator to improve estimation accuracy. The bound on the MSE in this case is derived by first averaging the conditional PDF over the nuisance parameters, and then evaluating the scalar bound on the parameter of interest. We refer to this approach as the *averaging approach* [77] which results in the *statistical channel knowledge bound*. The derivation of the bound depends on the specific channel model and may not be analytically tractable for channel models with arbitrary fading statistics. Therefore, we restrict our analysis to the widely used Ricean/Rayleigh channel models. Alternatively, one can adopt Monte Carlo techniques to approximate the result by considering a finite set of channel realizations.

A. Cramér-Rao Bound

We now derive the CRB taking into account the statistics of the CPRs at the receiver by first determining $f_{\mathbf{r}|\mathbf{v}}(\mathbf{r}|\mathbf{v}; \tau)$. Substituting (7) for a known $\mathbf{v}$ into (18), we obtain

$$\begin{aligned} f_{\mathbf{r}|\mathbf{v}}(\mathbf{r}|\mathbf{v}; \tau) &\propto \exp\left\{-\gamma_{\text{snr}}\left(\sum_{k=1}^L \sum_{l=1}^L (v_k v_l + v_{k+L} v_{l+L}) \rho_{\mathbf{g}}((k-l)\Delta\tau) \right. \right. \\ &\quad \left. \left. - 2 \sum_{k=1}^L v_k q_k^I(\tau) + v_{k+L} q_k^Q(\tau)\right)\right\} \end{aligned} \quad (35)$$

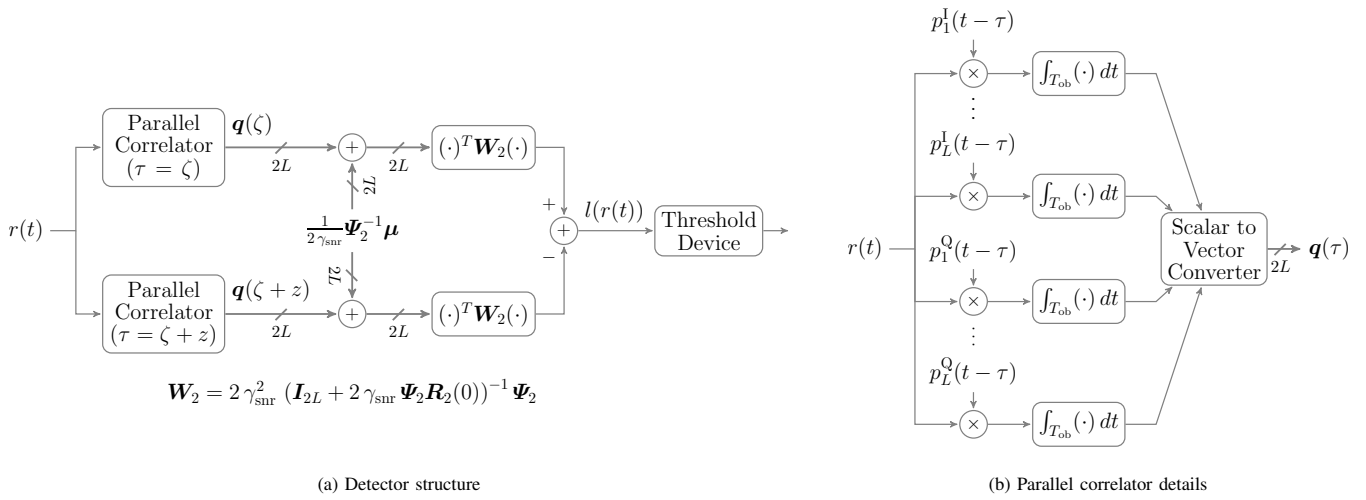


Fig. 1. Optimum detector in the presence of Ricean/Rayleigh fading with knowledge of the channel statistics at the receiver. The details of each parallel correlator in (a) are depicted in (b).

where $q_k^I(\tau)$ and $q_k^Q(\tau)$ are realizations of the random variables

$$q_k^I(\tau) \triangleq \frac{1}{E_p} \int_0^{T_{\text{ob}}} r(t) p_k^I(t - \tau) dt \quad (36)$$

$$q_k^Q(\tau) \triangleq \frac{1}{E_p} \int_0^{T_{\text{ob}}} r(t) p_k^Q(t - \tau) dt \quad (37)$$

and

$$p_k^I(t) \triangleq \sqrt{2E_p} g(t - (k-1)\Delta\tau) \cos(2\pi f_c t) \quad (38)$$

$$p_k^Q(t) \triangleq -\sqrt{2E_p} g(t - (k-1)\Delta\tau) \sin(2\pi f_c t). \quad (39)$$

For convenience we define the random vector $\mathbf{q}(\tau) = [q_1^I(\tau), \dots, q_L^I(\tau), q_1^Q(\tau), \dots, q_L^Q(\tau)]^T$. Now, the expression in (35) can be rewritten compactly as

$$f_{\mathbf{r}|\mathbf{v}}(\mathbf{r}|\mathbf{v}; \tau) \propto \exp \left\{ -\gamma_{\text{snr}} \text{tr} \left\{ \mathbf{R}_2(0) [\mathbf{v} - \mathbf{R}_2(0)^{-1} \mathbf{q}(\tau)] \right. \right. \\ \left. \left. \times [\mathbf{v} - \mathbf{R}_2(0)^{-1} \mathbf{q}(\tau)]^T - \mathbf{R}_2(0)^{-1} \mathbf{q}(\tau) \mathbf{q}(\tau)^T \right\} \right\}. \quad (40)$$

Note that $\mathbf{R}_2(0) = \mathbf{I}_2 \otimes \mathbf{R}(0)$ is a symmetric matrix.

According to the averaging approach, we can utilize the PDF $f_{\mathbf{r}}(\mathbf{r}|\tau)$ obtained by averaging the conditional PDF $f_{\mathbf{r}|\mathbf{v}}(\mathbf{r}|\mathbf{v}; \tau)$ over the nuisance parameters $\mathbf{v}$. We note that the quantity in the exponent of (40) is related to the non-central Wishart distribution [108], and thus the expression for $f_{\mathbf{r}}(\mathbf{r}|\tau)$ can be determined from the MGF of such a distribution. Applying expressions in [108], the PDF for Ricean/Rayleigh fading can be written as

$$f_{\mathbf{r}}(\mathbf{r}|\tau) \propto \frac{\exp \left\{ -\frac{1}{2} \text{tr} \left\{ \Psi_2^{-1} \mu \mu^T \right\} \right\}}{\det \left\{ \mathbf{I}_L + 2\gamma_{\text{snr}} \Psi_2 \mathbf{R}(0) \right\}} \\ \times \exp \left\{ 2\gamma_{\text{snr}}^2 \text{tr} \left\{ \left[\mathbf{I}_{2L} + 2\gamma_{\text{snr}} \Psi_2 \mathbf{R}_2 \right]^{-1} \Psi_2 \right. \right. \\ \left. \left. \times \left[\mathbf{q}(\tau) + \frac{\Psi_2^{-1} \mu}{2\gamma_{\text{snr}}} \right] \left[\mathbf{q}(\tau) + \frac{\Psi_2^{-1} \mu}{2\gamma_{\text{snr}}} \right]^T \right\} \right\}. \quad (41)$$

Starting from (41), it is shown in Appendix II that the CRB is given by

$$\eta_{\text{CR}} = \left[-\mathbb{E} \left\{ \frac{\partial^2}{\partial \tau^2} \ln f_{\mathbf{r}}(\mathbf{r}|\tau) \right\} \right]^{-1} \quad (42)$$

$$= \left[2\gamma_{\text{snr}} \mu^T \Upsilon \mu + 4\gamma_{\text{snr}}^2 \text{tr} \left\{ \left[\mathbf{I}_{2L} + 2\gamma_{\text{snr}} \Psi_2 \mathbf{R}_2(0) \right]^{-1} \Psi_2 \right. \right. \\ \left. \left. \times \left[\Upsilon \Psi_2 \mathbf{R}_2(0) - \Xi(\Psi_2 + \mu \mu^T) \Xi^T \right] \right\} \right]^{-1} \quad (43)$$

where the matrices Υ and Ξ are given in (21) and (61), respectively. Note that the closed form result in (43) is an expression for the CRB that takes into account the available a priori information about the multipath components, and hence serves as a lower bound on the TOA estimation error when the receiver has statistical knowledge of the CPR.

The expression in (43) can be further simplified for specific cases of interest. In particular, for resolvable multipath channels with Rayleigh fading, the expression reduces to:

$$\eta_{\text{CR}} = \left[32\pi^2 \beta_g^2 \gamma_{\text{snr}}^2 \sum_{l=1}^L \frac{\sigma_l^4}{1 + 2\sigma_l^2 \gamma_{\text{snr}}} \right]^{-1} \quad (44)$$

where $2\sigma_l^2$ is the power gain of the scattered component of the l th path defined below (10) in Sec. II. Note that under this condition, the CRB no longer depends on the center frequency f_c , but only on the signal bandwidth through β_g .

Conventional TOA estimation systems typically leverage only the detection of the first path, which is characterized by a SNR equal to γ_p , by neglecting the MPC. We note that for large γ_{snr} , the expression in (44) reduces to:

$$\eta_{\text{CR}} \approx \left[8\pi^2 \beta_g^2 \gamma_{\text{snr}} \right]^{-1} \quad (45)$$

where we have used $\sum_{l=1}^L 2\sigma_l^2 = 1$. Comparing (45) to the noncoherent version of (24), we observe that additional information contained within the remaining MPCs may be exploited for a potential performance gain of $\gamma_{\text{snr}}/\gamma_p$.

B. Ziv-Zakai Bound

In this section, the ZZB is derived where $\mathbf{v}$ is considered to be a random vector with distribution known at the receiver. As shown in Sec. III, the steps to determine the ZZB are to first design the optimum binary detector (15), evaluate its performance $P_{\min}(z)$, and then use $P_{\min}(z)$ in the evaluation of (16). By using (15) and substituting $\tau = \zeta$, for hypothesis H_1 , and $\tau = \zeta + z$, for hypothesis H_2 , in the right hand side of (41), the log-likelihood ratio (LLR) can be written as⁷

$$\ln \frac{f_{\mathbf{r}|\tau}(\mathbf{r}|\zeta)}{f_{\mathbf{r}|\tau}(\mathbf{r}|\zeta+z)} = \text{tr} \left\{ 2\gamma_{\text{snr}}^2 \left[\mathbf{I}_{2L} + 2\gamma_{\text{snr}} \boldsymbol{\Psi}_2 \mathbf{R}_2(0) \right]^{-1} \boldsymbol{\Psi}_2 \right. \\ \times \left(\left[\mathbf{q}(\zeta) + \frac{\boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu}}{2\gamma_{\text{snr}}} \right] \left[\mathbf{q}(\zeta) + \frac{\boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu}}{2\gamma_{\text{snr}}} \right]^T \right. \\ \left. \left. - \left[\mathbf{q}(\zeta+z) + \frac{\boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu}}{2\gamma_{\text{snr}}} \right] \left[\mathbf{q}(\zeta+z) + \frac{\boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu}}{2\gamma_{\text{snr}}} \right]^T \right) \right\}. \quad (46)$$

We denote the LLR (46) as $l(r(t))$ in order to emphasize its dependence on the observation $r(t)$ and to develop the optimal detector structure. Note that the LLR corresponds to the optimal binary PPM Rake receiver (correlator-squarer receiver), which is depicted in Fig. 1.⁸

The detector computes $l(r(t))$ and decides for hypothesis H_1 if $l(r(t)) > 0$; otherwise hypothesis H_2 is chosen. From (46), the decision statistic under hypothesis H_1 is

$$y = l(r(t)|H_1) = \text{tr} \{ \mathbf{S} \mathbf{U} \} \quad (47)$$

where $\mathbf{S}$ is a $4L \times 4L$ block diagonal matrix

$$\mathbf{S} \triangleq 2\gamma_{\text{snr}}^2 \begin{bmatrix} \mathbf{I}_2 & \mathbf{0} \\ \mathbf{0} & -\mathbf{I}_2 \end{bmatrix} \otimes \left[\mathbf{I}_L + 2\gamma_{\text{snr}} \boldsymbol{\Psi} \mathbf{R}(0) \right]^{-1} \boldsymbol{\Psi} \quad (48)$$

and

$$\mathbf{U} = \begin{bmatrix} \mathbf{q}^{(1)} + (2\gamma_{\text{snr}})^{-1} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \\ \mathbf{q}^{(2)} + (2\gamma_{\text{snr}})^{-1} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \end{bmatrix} \quad (49)$$

where $\mathbf{q}^{(1)} \triangleq \mathbf{q}(\zeta)|_{H_1}$ and $\mathbf{q}^{(2)} \triangleq \mathbf{q}(\zeta+z)|_{H_1}$. For further convenience, we express the random matrix $\mathbf{U}$ as the result of a zero mean random vector plus a deterministic component: $\mathbf{U} = (\mathbf{x} + \mathbf{m})(\mathbf{x} + \mathbf{m})^T$, where the vector $\mathbf{x}$ is a $4L \times 1$ zero mean random vector given by

$$\mathbf{x} \triangleq \begin{bmatrix} \mathbf{q}^{(1)} - \mathbb{E} \{ \mathbf{q}^{(1)} \} \\ \mathbf{q}^{(2)} - \mathbb{E} \{ \mathbf{q}^{(2)} \} \end{bmatrix} \quad (50)$$

and the deterministic vector $\mathbf{m}$ is given by

$$\mathbf{m} = \begin{bmatrix} \mathbb{E} \{ \mathbf{q}^{(1)} \} + (2\gamma_{\text{snr}})^{-1} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \\ \mathbb{E} \{ \mathbf{q}^{(2)} \} + (2\gamma_{\text{snr}})^{-1} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \end{bmatrix} \quad (51)$$

⁷According to the averaging approach, we have used the PDF $f_{\mathbf{r}|\tau}(\mathbf{r}|\tau)$ obtained by averaging the conditional PDF $f_{\mathbf{r}|\mathbf{v},\tau}(\mathbf{r}|\mathbf{v},\tau)$ over the nuisance parameters $\mathbf{v}$.

⁸Note that to the best of the authors' knowledge both the structure and performance of this receiver have not been investigated in the literature and they can be considered as a byproduct of this paper.

where the expressions for the quantities $\mathbb{E} \{ \mathbf{q}^{(1)} \}$ and $\mathbb{E} \{ \mathbf{q}^{(2)} \}$ are given in Appendix III.

The probability of error $P_{\min}(z)$, required to evaluate the ZZB in (16), is given by

$$P_{\min}(z) = \frac{1}{2} P_{e|H_1}(z) + \frac{1}{2} P_{e|H_2}(z) \quad (52)$$

where $P_{e|H_1}(z)$ and $P_{e|H_2}(z)$ are the error probabilities conditioned on hypotheses H_1 and H_2 , respectively. Due to symmetry, $P_{e|H_1}(z) = P_{e|H_2}(z)$, and hence it is sufficient to evaluate $P_{e|H_1}(z)$. Thus, $P_{\min}(z)$ is given by

$$P_{\min}(z) = P_{e|H_1}(z) = \mathbb{P} \{ y < 0 | H_1 \}. \quad (53)$$

To evaluate (53) it is necessary to characterize the statistics of the RV y . It follows from Appendix III that $\mathbf{x}$ is a zero-mean normal random vector with covariance matrix $\boldsymbol{\Sigma}$ given in (80). Therefore $\mathbf{U}$ follows a noncentral Wishart distribution with one degree of freedom. In order to evaluate (53), we take the approach of first deriving the MGF of y . Using expressions in [108], the MGF is given by

$$\Phi_y(\xi) = \frac{\exp \left\{ \text{tr} \left\{ \xi \mathbf{S} [\mathbf{I}_{4L} - 2\xi \boldsymbol{\Sigma} \mathbf{S}]^{-1} \mathbf{M} \right\} \right\}}{\det \{ \mathbf{I}_{4L} - 2\xi \boldsymbol{\Sigma} \mathbf{S} \}^{1/2}} \quad (54)$$

where $\mathbf{M} = \mathbf{m} \mathbf{m}^T$. Details on the terms in (54) are given in Appendix III.

From (81) we can see that the denominator in (54) does not have any dependence on f_c as $\mathbf{R}(z)$ depends on only the baseband pulse $g(t)$. Furthermore, in the case of purely Rayleigh fading, $\mathbf{m} = \mathbf{0}$ and the numerator in (54) becomes one, thus (54) has no dependence on the center frequency. Therefore, we would expect the performance to approach that of the noncoherent case.

Several techniques are available to evaluate (53) using the MGF (54). The classical way is to use the Inversion Theorem [111] and compute the following integral numerically

$$P_{\min}(z) = \mathbb{P} \{ y < 0 \} = \frac{1}{2} - \frac{1}{\pi} \Re \left\{ \int_0^\infty \frac{\Phi_y(jv)}{jv} dv \right\} \\ = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\Im \{ \Phi_y(jv) \}}{v} dv. \quad (55)$$

VI. NUMERICAL RESULTS

In this section, numerical illustrations of the results are provided. Unless otherwise stated, a bandpass root raised cosine (RRC) pulse with center frequency $f_c = 4$ GHz, roll-off $\nu = 0.6$, and bandwidth $W = 1.6$ GHz is considered [100]. It follows that $\Delta\tau = 1/W = 0.62$ ns. The unknown time delay τ is considered uniformly distributed in $[0, T_a)$ with $T_a = 100$ ns.

A. Results with Perfect Knowledge of the CPR: Conditioning Approach

We begin by considering the conditioning approach given in Sec. IV. In Figs. 2 and 3, the ZZB and the CRB/MCRB on the RMSE are plotted as a function of the total SNR for uniform power dispersion profile (PDP) ($\Gamma_l = 1/L$ for each l). Different multipath conditions are considered assuming the receiver has perfect knowledge of the CPR. The MCRBs for

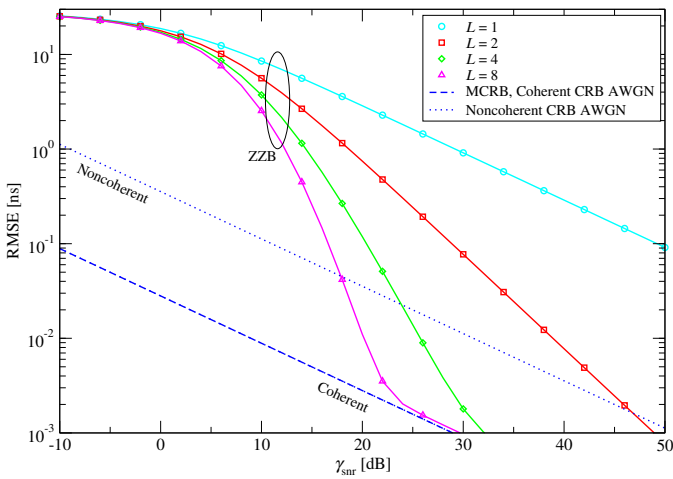


Fig. 2. ZZB and CRB/MCRB on the RMSE, using the conditioning approach, as a function of SNR for the Rayleigh fading channel with $L = 1, 2, 4, 8$. The CRB is also shown for coherent and noncoherent estimation in AWGN. Note that for coherent estimation, the curves of the MCRB for the fading channel coincide with that of the CRB for AWGN (blue dashed line).

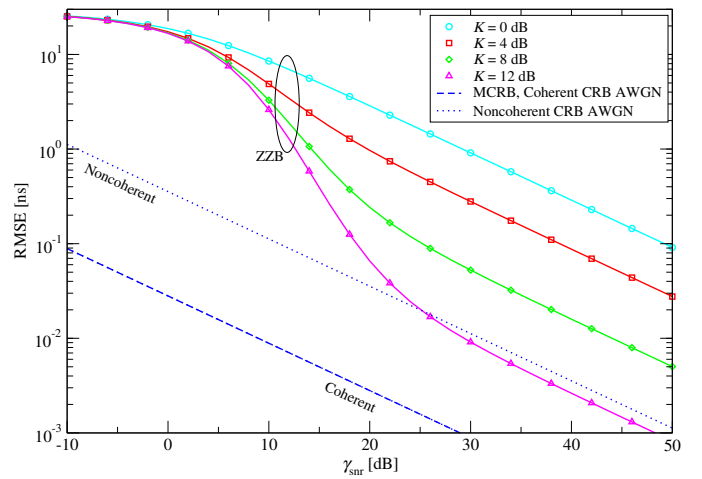


Fig. 3. ZZB and CRB/MCRB on the RMSE, using the conditioning approach, as a function of SNR for the Ricean channel with $L = 1$ and $K = 0, 4, 8, 12$ dB. The CRB is also shown for coherent and noncoherent estimation in AWGN. Note that for coherent estimation, the curves of the MCRB for the fading channel coincide with that of the CRB for AWGN (blue dashed line).

the channel models of interest are generated by evaluating (22), while the ZZBs are evaluated using (16) in conjunction with (32) and (33).⁹

The presence of the threshold effect can be seen from the ZZB curves in Figs. 2 and 3. In fact, the a priori region is clearly indicated by the ZZB curves, which approach $T_a/\sqrt{12}$ for low SNR values. However, the CRB does not account for this behavior. For comparison purposes the CRB for noncoherent estimation in AWGN is also reported. The noncoherent bound is significantly higher than the coherent one because of phase uncertainty, as discussed in Sec. IV-A.

In particular, Fig. 2 depicts the performance, in terms of RMSE, for Rayleigh fading channels with $L = 1, 2, 4$, and 8 paths.¹⁰ In this figure, we observe that the performance for the ZZB converges to the coherent MCRB at high SNRs. This is due to the fact that under the conditioning approach, perfect channel knowledge is available at the receiver. As shown later, this will not be the case for the averaging approach. We further note that this convergence happens more quickly as L increases. This can be ascribed to the fact that when L increases the same amount of energy is spread over different paths subject to independent fading. The result is a diversity gain for medium SNR values (note that the slope of the curve is $-L$ in logarithmic scale), which vanishes at high SNRs. This is in contrast to diversity behavior in communication systems where the gain is maintained at high SNRs.

In Fig. 3 we depict the results for single-path Ricean

channels with Ricean factor values of $K = 0, 4, 8, 12$ dB. We see that the ZZB tends toward the coherent MCRB, however, unlike the Rayleigh fading case, there is a significant gap between the two bounds. This shows that the MCRB is an optimistic bound, even for a low level of fading (large K), especially at low SNRs.

The plots in both Figs. 2 and 3 show that the MCRBs for each value of L and/or K are identical. This is in agreement with the results of Sec. IV-A, where it was shown that the MCRB is only dependent on the effective bandwidth of the transmitted pulse in Rayleigh fading channels or Ricean/Rayleigh fading channels where only one path is affected by Ricean fading. Furthermore, the MCRB is essentially equivalent to the coherent CRB for AWGN, since, under the conditioning approach the receiver has knowledge of all the MPCs. However, the comparison between the ZZB and MCRB reveals that the ZZB is more capable of capturing the effect of the channel, e.g., diversity behavior, thus resulting in a tighter bound.

B. Results with Statistical Knowledge of the CPR: Averaging Approach

We now consider the averaging approach given in Sec. V.

In Figs. 4–8, the ZZB and the CRB on the RMSE are plotted as a function of the SNR for uniform PDP ($\Gamma_l = 1/L$ for each l). Different multipath conditions are considered assuming the receiver has a priori statistical knowledge of the CPR. The CRBs for the channel models of interest are generated by evaluating (43) and (44), while the ZZBs are evaluated using (16) in conjunction with (54) and (55).

Fig. 4 illustrates the performance for Rayleigh fading channels with $L = 1, 2, 4$, and 8 paths. Curves for the CRB, plotted using (43) and shown by dashed curves, indicate that the RMSE is higher than that of the noncoherent AWGN case. This is to be expected since the CRB curves given by

⁹In a strict sense, the CRB and ZZB plots may not be directly comparable as they refer to estimation under different scenarios (deterministic unknown (non-Bayesian) parameter vs. random unknown (Bayesian) parameter). Unfortunately, the Bayesian CRB is not available with a uniform prior on the parameter due to violation of the regularity conditions on the PDF. While a uniform distribution is assumed for the prior on τ for the ZZB, it is usual to adopt the non-Bayesian CRB for comparison purposes [66]. Thus, only values of the CRB much smaller than the initial uncertainty of τ should be considered when comparing.

¹⁰Note that the case $L = 1$ does not correspond to the AWGN channel as the path is still subject to fading.

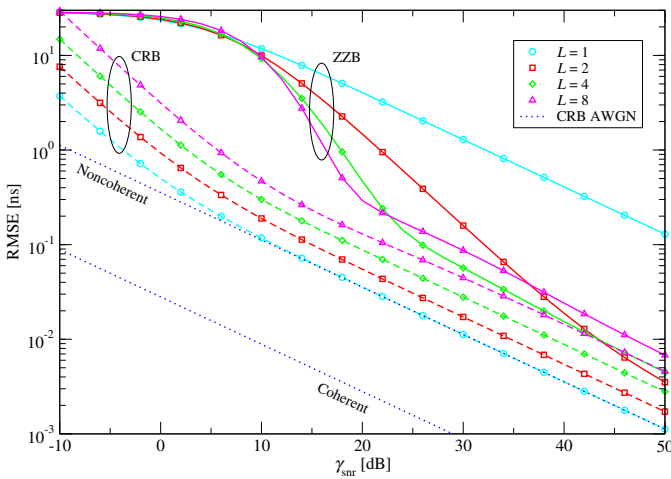


Fig. 4. ZZB and CRB on the RMSE, using the averaging approach, as a function of SNR for the Rayleigh fading channel with $L = 1, 2, 4, 8$. The CRB is also shown for coherent and noncoherent estimation in AWGN. The plot shows the CRB based on the general expression (43) with dashed line style.

(43) incorporate the effect of several factors, including: the random amplitude and phase of the paths, and the effect of pulse overlap between adjacent paths.

From Fig. 4, we observe that the behavior of the ZZB curves greatly varies depending on L . In fact, for a large number of paths L , the ZZB decreases more quickly (diversity gain) from the a priori region at low SNRs. However, at high SNRs, larger L does not always yield better performance due to the increased dispersion of the CPR. In this case the increased diversity gain is not sufficient to counteract the dispersion, which includes the spreading of energy among MPCs and the additional occurrence of pulse overlap from adjacent paths. To better understand the effect of energy spread, we plot the RMSE as a function of the path SNR (γ_p) in Fig. 5, which effectively shifts the curves in Fig. 4 to maintain consistent path SNR for different values of L . Comparison of Figs. 4 and 5 shows that the crossing of the ZZB curves in Fig. 4 is largely due to reduction in path SNR as the number of paths increases. These figures also show that, as expected, the ZZB is higher than the CRB for Rayleigh fading.

Comparing Figs. 4 and 5 shows that there is some degradation in performance for the CRB as the number of paths increase. Clearly the majority of degradation is due to the decrease in first-path SNR, as evidenced by comparing the two figures. However, even when the paths have consistent SNR, as in Fig. 5, we see that only the $L = 1$ case converges to the CRB for noncoherent estimation in AWGN. The degradation in cases with $L > 1$ is caused by pulse overlap. To further explore this behavior, we take a closer look at the CRB in Fig. 6 as a function of total SNR for Rayleigh fading channels with $L = 2$ and $L = 8$ having different path spacing $\Delta\tau$. Note that as $\Delta\tau$ increases, the difference between the CRBs for $L = 2$ and $L = 8$ decreases and the CRBs approach the noncoherent CRB for AWGN at high SNRs. These asymptotic behaviors are accurately predicted by (44) for the case of no pulse overlap and by (45) for the case of no pulse overlap

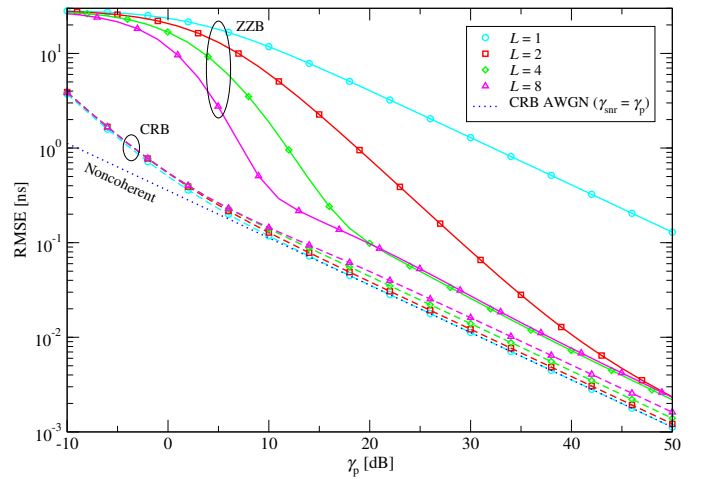


Fig. 5. ZZB and CRB on the RMSE, using the averaging approach, as a function of path SNR for the Rayleigh fading channel with $L = 1, 2, 4, 8$. The CRB is also shown for noncoherent estimation in AWGN for the case $\gamma_{\text{snr}} = \gamma_p$. The plot shows the CRB based on the general expression (43) with dashed line style.

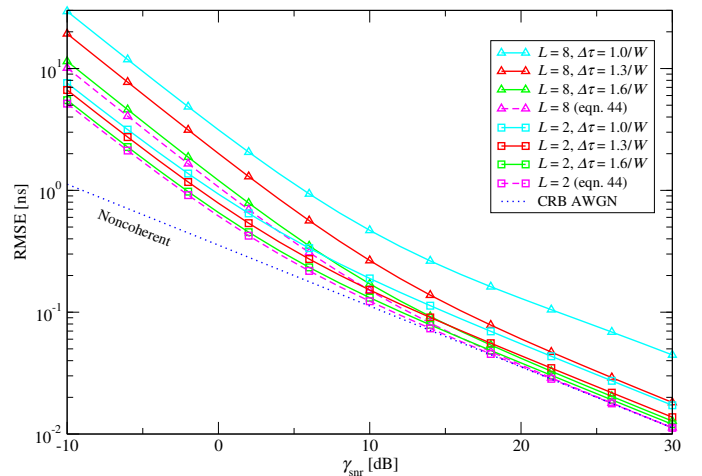


Fig. 6. CRB on the RMSE, using the averaging approach, as a function of total SNR for the Rayleigh fading channel with $L = 2, 8$. The CRB is also shown for noncoherent estimation in AWGN. The plot shows both the CRB based on the general expression (43) and for the resolvable multipath case (44) with solid and dashed line styles, respectively. Note that as the path separation in the channel increases, the performance approaches the resolvable case.

with high SNR. On the other hand, (43) is a more general expression that is capable of capturing the subtle effects of different channel and path configurations on the CRB.

To illustrate the effect of the specular component in the path gains on the RMSE, we first investigate the single-path case. Fig. 7 shows the CRB and ZZB on the RMSE for the case of a single-path Ricean channel, as described in Sec. II, for various values of Ricean factor, $K = 0, 4, 8, 12$ dB. The figure indicates that the CRB transitions from the coherent CRB for AWGN at low SNRs to the noncoherent CRB for AWGN at high SNRs. The figure also indicates that for larger values of K the ZZB decreases more quickly from the a priori region toward the associated CRB. This occurs because larger K means that there is a reduced amount of fading and phase uncertainty in the single-path channel, which in turn improves

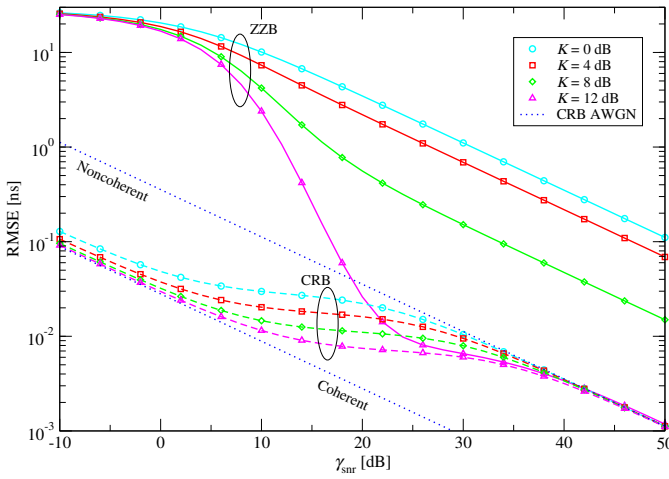


Fig. 7. ZZB and CRB, using the averaging approach, on the RMSE as a function of SNR for the single-path Ricean channel with $K = 0, 4, 8, 12$ dB. The CRB is also shown for coherent and noncoherent estimation in AWGN.

the accuracy of TOA estimation. However, convergence of the ZZB to the CRB occurs only for sufficiently large values of K . We believe this is because given the fading statistics it can still be difficult to overcome the a priori uncertainty caused by T_{ob} . Depending on the value of K , the ZZB decreases below the noncoherent CRB for AWGN. In fact, for $K = 12$ dB we see this behavior for SNRs greater than approximately 20 dB.

Fig. 8 considers channels where the first path is affected by Ricean fading and the remaining paths experience Rayleigh fading. Here we consider a Ricean factor of $K = 10$ dB. Compared to the purely Rayleigh fading case (Fig. 4), the effect of L is less pronounced. This is due to the presence of a specular component in the first path that carries most of the TOA information, and thus causes the ZZB to decrease more rapidly regardless of the number of paths. However, we note that larger L causes the ZZB to converge more closely to the associated CRB at high SNRs. Again, we see that larger values of L do not always imply improved performance for all values of SNR. For example, at $\gamma_{\text{snr}} = 25$ dB the performance for $L = 8$ is better than $L = 1$, but worse than $L = 2$ and $L = 4$. This can be attributed to the dispersion of the CPRs, which includes differences in the individual path powers and the effect of pulse overlap, as remarked before.

Nakagami- m fading, which has been extensively adopted to characterize UWB channels, is often well approximated as a Ricean distribution with a suitable mapping of parameters. Given the Nakagami parameters $m \geq 1$ and Γ , the Ricean factor K and σ can be evaluated through the following relationships [100]

$$K = \frac{\sqrt{m^2 - m}}{m - \sqrt{m^2 - m}}, \quad \sigma^2 = \frac{\Gamma}{2(K + 1)}. \quad (56)$$

In Fig. 9 we consider a multipath channel with exponential PDP with average path power gains given by

$$\Gamma_l = \frac{[\exp\{\Delta/\varepsilon_\Gamma\} - 1] \exp\{-\Delta(l-1)/\varepsilon_\Gamma\}}{\exp\{\Delta/\varepsilon_\Gamma\} [1 - \exp\{-L\Delta/\varepsilon_\Gamma\}]} \quad (57)$$

and Nakagami severity parameter $m_l = m_1 \exp\{-(l-1)/\varepsilon_m\}$ for $l = 1, 2, \dots, L$ [57], [105] using the Ricean approximation

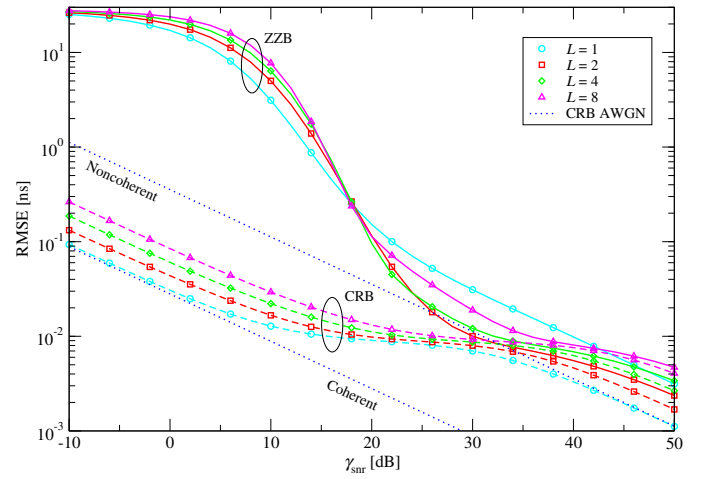


Fig. 8. ZZB and CRB, using the averaging approach, on the RMSE as a function of SNR for the Ricean/Rayleigh channel. Ricean factor $K = 10$ dB is considered with $L = 1, 2, 4, 8$. In this channel model the first path is Ricean distributed and the remaining $L - 1$ paths are Rayleigh distributed. The CRB is also shown for coherent and noncoherent estimation in AWGN.

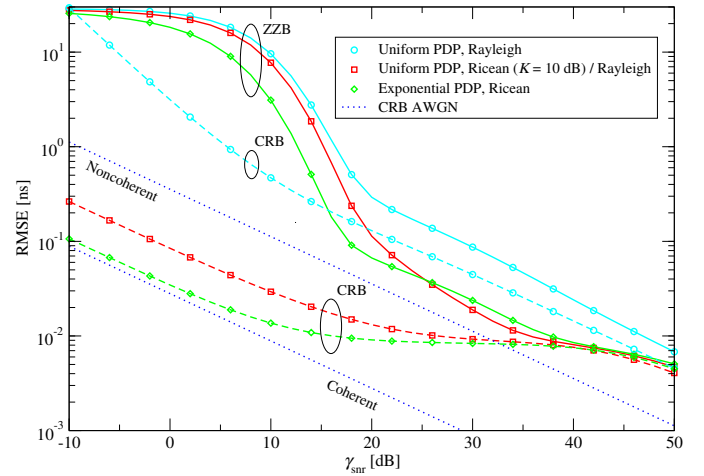


Fig. 9. ZZB and CRB, using the averaging approach, on the RMSE as a function of SNR for different PDPs and channels with $L = 8$. The CRB is also shown for coherent and noncoherent estimation in AWGN.

according to (56). The parameter ε_Γ describes the multipath spread of the channel. We consider $L = 8$, $\varepsilon_\Gamma = 6$ ns, $\varepsilon_m = 4$, and $m_1 = 3$. The comparison between different channel models is reported in Fig. 9 for the case of: 1) uniform PDP with Rayleigh distributed paths, 2) uniform PDP with the first path Ricean distributed ($K = 10$ dB) and the remaining paths Rayleigh distributed, and 3) exponential PDP according to (57) with Ricean distributed paths. From low to moderate SNRs, the best performance is given by the exponential PDP. In fact with the exponential PDP, most of the energy is concentrated close to the first path; therefore, when a MPC is erroneously detected as the first path, it is more likely that it is close to the correct one. For higher SNRs the best performance is essentially a tie between the exponential PDP and uniform PDP with Ricean/Rayleigh fading.

VII. CONCLUSION

Analytical expressions for the CRB and ZZB on the RMSE of TOA estimation for wideband and UWB ranging systems operating in realistic multipath environments are provided. Under different a priori multipath information, it is shown that the classical CRB may not account for the effect of multipath properly in the practical SNR regime, whereas the ZZB provides improved bounds. The analysis of the ZZB reveals how information about the CPR can significantly improve the estimation performance, especially in a Ricean or a rich multipath channel. The intertwined effect of diversity and CPR dispersion in ranging systems is investigated showing that, contrary to communication systems, diversity behavior is not exhibited in the asymptotically large SNR regime, but rather at medium SNRs. The bounds investigated in this paper serve as useful benchmarks for the design of improved TOA estimators that exploit multipath.

APPENDIX I

COMMENTS ON CRAMÉR RAO BOUND WITH PERFECT CPR KNOWLEDGE

In some cases listed in Sec. IV-A, the bound in (22) reduces to the one in the AWGN case. This can be seen as follows. In the first case, when $\boldsymbol{\mu} = \mathbf{0}$, the second term in the denominator of (22) becomes zero and $\text{tr}\{\boldsymbol{\Psi}\} = \sum_{l=1}^{2L} \sigma_l^2 = 1$, due to normalization. For the second and third cases, expanding the second term in the denominator using (21) gives:

$$\begin{aligned} \boldsymbol{\mu}^T \boldsymbol{\Upsilon} \boldsymbol{\mu} &= (2\pi f_c)^2 \left[\boldsymbol{\mu}_1^T \mathbf{R}(0) \boldsymbol{\mu}_1 + \boldsymbol{\mu}_2^T \mathbf{R}(0) \boldsymbol{\mu}_2 \right] + \boldsymbol{\mu}_1^T \dot{\mathbf{R}} \boldsymbol{\mu}_1 \\ &\quad + \boldsymbol{\mu}_2^T \dot{\mathbf{R}} \boldsymbol{\mu}_2 + 4\pi f_c \left[\boldsymbol{\mu}_1^T \mathbf{R}_{g\dot{g}} \boldsymbol{\mu}_2 - \boldsymbol{\mu}_2^T \mathbf{R}_{g\dot{g}} \boldsymbol{\mu}_1 \right] \end{aligned}$$

where

$$\begin{aligned} \boldsymbol{\mu}_1 &= [\mu_1, \mu_2, \dots, \mu_L]^T \\ \boldsymbol{\mu}_2 &= [\mu_{L+1}, \mu_{L+2}, \dots, \mu_{2L}]^T. \end{aligned}$$

For the case when μ_l and μ_{L+l} are non-zero for only one $\tilde{l} \in \{1, 2, \dots, L\}$, since $\mathbf{R}_{g\dot{g}}$ is skew-symmetric, it can be shown that

$$\begin{aligned} \boldsymbol{\mu}^T \boldsymbol{\Upsilon} \boldsymbol{\mu} &= (2\pi f_c)^2 (\mu_{\tilde{l}}^2 + \mu_{L+\tilde{l}}^2) [\mathbf{R}(0)]_{\tilde{l}\tilde{l}} + (\mu_{\tilde{l}}^2 + \mu_{L+\tilde{l}}^2) [\dot{\mathbf{R}}]_{\tilde{l}\tilde{l}} \\ &= (2\pi)^2 \beta_g^2 (\mu_{\tilde{l}}^2 + \mu_{L+\tilde{l}}^2) \end{aligned}$$

where we have used the fact that $[\mathbf{R}(0)]_{\tilde{l}\tilde{l}} = 1$ and $[\dot{\mathbf{R}}]_{\tilde{l}\tilde{l}} = (2\pi)^2 \beta_g^2$. Finally, when adjacent pulses are resolvable it can be shown that $\mathbf{R}(0) = \mathbf{I}_L$, $\dot{\mathbf{R}} = (2\pi)^2 \beta_g^2 \mathbf{I}_L$, and $\mathbf{R}_{g\dot{g}} = \mathbf{0}$. Thus, in this case for arbitrary $\boldsymbol{\mu}$, $\boldsymbol{\mu}^T \boldsymbol{\Upsilon} \boldsymbol{\mu} = (2\pi)^2 (f_c^2 + \beta_g^2) \boldsymbol{\mu}^T \boldsymbol{\mu}$.

APPENDIX II

CRAMÉR RAO BOUND USING THE AVERAGING APPROACH

In this Appendix we derive the expression of the CRB given in (43). Note that the statistical expectation in (42) is taken with respect to $\mathbf{r}$, which is embedded in $\mathbf{q}(\tau)$, whose elements

are defined in (36) and (37). Their first and second derivatives are, respectively,

$$\dot{\mathbf{q}}(\tau)_k = \begin{cases} \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{r}(t) \frac{\partial}{\partial \tau} p_k^I(t - \tau) dt & 1 \leq k \leq L \\ \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{r}(t) \frac{\partial}{\partial \tau} p_{k-L}^Q(t - \tau) dt & L < k \leq 2L \end{cases} \quad (58)$$

$$\ddot{\mathbf{q}}(\tau)_k = \begin{cases} \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{r}(t) \frac{\partial^2}{\partial \tau^2} p_k^I(t - \tau) dt & 1 \leq k \leq L \\ \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{r}(t) \frac{\partial^2}{\partial \tau^2} p_{k-L}^Q(t - \tau) dt & L < k \leq 2L. \end{cases} \quad (59)$$

In particular, we can write

$$\begin{aligned} & -\mathbb{E} \left\{ \frac{\partial^2}{\partial \tau^2} \ln f_{\mathbf{r}}(\mathbf{r}|\tau) \right\} \\ &= -4 \gamma_{\text{snr}}^2 \text{tr} \left\{ [\mathbf{I}_{2L} + 2 \gamma_{\text{snr}} \boldsymbol{\Psi}_2 \mathbf{R}_2(0)]^{-1} \boldsymbol{\Psi}_2 \right. \\ &\quad \times \mathbb{E} \left\{ \ddot{\mathbf{q}}(\tau) \left[\mathbf{q}(\tau) + \frac{1}{2 \gamma_{\text{snr}}} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \right]^T + \dot{\mathbf{q}}(\tau) \dot{\mathbf{q}}(\tau)^T \right\} \left. \right\}. \end{aligned} \quad (60)$$

It can then be shown that

$$\begin{aligned} \mathbb{E} \{ \ddot{\mathbf{q}}(\tau) \mathbf{q}(\tau)^T \} &= -\boldsymbol{\Upsilon} \left[\mathbb{E} \{ \mathbf{v} \mathbf{v}^T \} \mathbf{R}_2(0) + \frac{1}{2 \gamma_{\text{snr}}} \mathbf{I}_{2L} \right] \\ \mathbb{E} \{ \ddot{\mathbf{q}}(\tau) \} &= -\boldsymbol{\Upsilon} \boldsymbol{\mu} \\ \mathbb{E} \{ \dot{\mathbf{q}}(\tau) \dot{\mathbf{q}}(\tau)^T \} &= \boldsymbol{\Xi} \mathbb{E} \{ \mathbf{v} \mathbf{v}^T \} \boldsymbol{\Xi}^T + \frac{1}{2 \gamma_{\text{snr}}} \boldsymbol{\Upsilon} \end{aligned}$$

where

$$\boldsymbol{\Xi} = \begin{bmatrix} \mathbf{R}_{g\dot{g}} & -2\pi f_c \mathbf{R}(0) \\ 2\pi f_c \mathbf{R}(0) & \mathbf{R}_{g\dot{g}} \end{bmatrix}. \quad (61)$$

Using the above expressions, the expectation in (60) reduces to

$$\begin{aligned} & \mathbb{E} \left\{ \ddot{\mathbf{q}}(\tau) \left[\mathbf{q}(\tau) + \frac{1}{2 \gamma_{\text{snr}}} \boldsymbol{\Psi}_2^{-1} \boldsymbol{\mu} \right]^T + \dot{\mathbf{q}}(\tau) \dot{\mathbf{q}}(\tau)^T \right\} \\ &= -\boldsymbol{\Upsilon} \boldsymbol{\Psi}_2 \mathbf{R}_2(0) - \boldsymbol{\Upsilon} \boldsymbol{\mu} \boldsymbol{\mu}^T \left[\mathbf{R}_2(0) + \frac{1}{2 \gamma_{\text{snr}}} \boldsymbol{\Psi}_2^{-1} \right] \\ &\quad + \boldsymbol{\Xi} (\boldsymbol{\Psi}_2 + \boldsymbol{\mu} \boldsymbol{\mu}^T) \boldsymbol{\Xi}^T. \end{aligned} \quad (62)$$

Inserting (62) into (60) gives (43).

APPENDIX III

DERIVATION OF THE STATISTICS OF $\mathbf{q}^{(1)}$ AND $\mathbf{q}^{(2)}$

This Appendix derives the statistical properties of $\mathbf{q}^{(1)}$ and $\mathbf{q}^{(2)}$ used in the derivation of the averaging approach of Sec. V. Under hypothesis H_1 , the expressions in (36) and (37) imply that

$$\begin{aligned} q_k^I(\tau + z) &= \sum_{m=1}^L \left(v_m \rho_g(z + (k-m)\Delta\tau) \cos(2\pi f_c z) \right. \\ &\quad \left. - v_{L+m} \rho_g(z + (k-m)\Delta\tau) \sin(2\pi f_c z) \right) \\ &\quad + n_k^I(\tau + z) \end{aligned} \quad (63)$$

$$\begin{aligned} q_k^Q(\tau + z) &= \sum_{m=1}^L \left(v_m \rho_g(z + (k-m)\Delta\tau) \sin(2\pi f_c z) \right. \\ &\quad \left. + v_{L+m} \rho_g(z + (k-m)\Delta\tau) \cos(2\pi f_c z) \right) \\ &\quad + n_k^Q(\tau + z) \end{aligned} \quad (64)$$

where we have used (5), (7), (38), and (39), and we have neglected the contribution of any double frequency terms. Since, for practical systems $T_{\text{ob}} f_c \gg 1$, without loss of generality one can assume that T_{ob} is chosen to be a multiple $k \gg 1$ of the carrier period $1/f_c$ so that any double frequency term would give no contribution when integrated over T_{ob} . Here the noise terms are given by

$$\mathbf{n}_k^I(\tau + z) = \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{n}(t) p_k^I(t - \tau - z) dt \quad (65)$$

$$\mathbf{n}_k^Q(\tau + z) = \frac{1}{E_p} \int_0^{T_{\text{ob}}} \mathbf{n}(t) p_k^Q(t - \tau - z) dt. \quad (66)$$

Note that $\mathbf{n}_k^I(\cdot)$ and $\mathbf{n}_k^Q(\cdot)$ are zero-mean Gaussian, but are, in general, correlated:

$$\begin{aligned} \mathbb{E} \{ \mathbf{n}_k^I(\tau + z_1) \mathbf{n}_l^I(\tau + z_2) \} &= \mathbb{E} \{ \mathbf{n}_k^Q(\tau + z_1) \mathbf{n}_l^Q(\tau + z_2) \} \\ &= \frac{1}{2\gamma_{\text{snr}}} \rho_g(z_2 - z_1 + (l - k)\Delta\tau) \cos(2\pi f_c(z_2 - z_1)) \end{aligned} \quad (67)$$

$$\begin{aligned} \mathbb{E} \{ \mathbf{n}_k^I(\tau + z_1) \mathbf{n}_l^Q(\tau + z_2) \} &= -\mathbb{E} \{ \mathbf{n}_k^Q(\tau + z_1) \mathbf{n}_l^I(\tau + z_2) \} \\ &= \frac{1}{2\gamma_{\text{snr}}} \rho_g(z_2 - z_1 + (l - k)\Delta\tau) \sin(2\pi f_c(z_2 - z_1)). \end{aligned} \quad (68)$$

Now, we can write

$$\mathbf{q}^{(1)} = \mathbf{R}_2(0)^T \mathbf{v} + \mathbf{n}^{(1)} \quad (69)$$

$$\mathbf{q}^{(2)} = \mathbf{R}_2(z)^T \mathbf{v} + \mathbf{n}^{(2)} \quad (70)$$

where

$$\begin{aligned} \mathbf{n}^{(1)} &= [\mathbf{n}_1^I(\tau), \dots, \mathbf{n}_L^I(\tau), \mathbf{n}_{L+1}^Q(\tau), \mathbf{n}_{L+2}^Q(\tau), \dots, \mathbf{n}_{2L}^Q(\tau)]^T \\ \mathbf{n}^{(2)} &= [\mathbf{n}_1^I(\tau + z), \dots, \mathbf{n}_L^I(\tau + z), \\ &\quad \mathbf{n}_{L+1}^Q(\tau + z), \mathbf{n}_{L+2}^Q(\tau + z), \dots, \mathbf{n}_{2L}^Q(\tau + z)]^T. \end{aligned}$$

Under the Ricean/Rayleigh fading model of Sec. II, $\mathbf{v}$ is a Gaussian random vector; therefore, we conclude that $\mathbf{q}^{(1)}$ and $\mathbf{q}^{(2)}$ are jointly Gaussian random vectors. Using (69)–(70), we have

$$\mathbb{E} \{ \mathbf{q}^{(1)} \} = \mathbf{R}_2(0)^T \boldsymbol{\mu} \quad (71)$$

$$\mathbb{E} \{ \mathbf{q}^{(2)} \} = \mathbf{R}_2(z)^T \boldsymbol{\mu}. \quad (72)$$

Similarly, using (67)–(70), the covariances between $\mathbf{q}^{(1)}$ and $\mathbf{q}^{(2)}$ can be determined as

$$\begin{aligned} \mathbb{Cov} \{ \mathbf{q}^{(1)}, \mathbf{q}^{(1)} \} &= \mathbf{R}_2(0)^T \boldsymbol{\Psi}_2 \mathbf{R}_2(0) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}_2(0) \\ &= \mathbf{I}_2 \otimes \mathbf{A} \end{aligned} \quad (73)$$

$$\begin{aligned} \mathbb{Cov} \{ \mathbf{q}^{(1)}, \mathbf{q}^{(2)} \} &= \mathbb{Cov} \{ \mathbf{q}^{(2)}, \mathbf{q}^{(1)} \}^T \\ &= \mathbf{R}_2(0)^T \boldsymbol{\Psi}_2 \mathbf{R}_2(z) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}_2(z) \\ &= \boldsymbol{\Omega} \otimes \mathbf{F} \end{aligned} \quad (74)$$

$$\begin{aligned} \mathbb{Cov} \{ \mathbf{q}^{(2)}, \mathbf{q}^{(2)} \} &= \mathbf{R}_2(z)^T \boldsymbol{\Psi}_2 \mathbf{R}_2(z) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}_2(0) \\ &= \mathbf{I}_2 \otimes \mathbf{D} \end{aligned} \quad (75)$$

where

$$\mathbf{A} = \mathbf{R}(0)^T \boldsymbol{\Psi} \mathbf{R}(0) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}(0) \quad (76)$$

$$\mathbf{F} = \mathbf{R}(0)^T \boldsymbol{\Psi} \mathbf{R}(z) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}(z) \quad (77)$$

$$\mathbf{D} = \mathbf{R}(z)^T \boldsymbol{\Psi} \mathbf{R}(z) + \frac{1}{2\gamma_{\text{snr}}} \mathbf{R}(0) \quad (78)$$

$$\boldsymbol{\Omega} = \begin{bmatrix} \cos(2\pi f_c z) & \sin(2\pi f_c z) \\ -\sin(2\pi f_c z) & \cos(2\pi f_c z) \end{bmatrix}. \quad (79)$$

Using the results of (73)–(75), the covariance matrix of $\mathbf{x}$ in (50) can now be written as

$$\boldsymbol{\Sigma} = \mathbb{Cov} \{ \mathbf{x}, \mathbf{x} \} = \begin{bmatrix} \mathbf{I}_2 \otimes \mathbf{A} & \boldsymbol{\Omega} \otimes \mathbf{F} \\ \boldsymbol{\Omega}^T \otimes \mathbf{F}^T & \mathbf{I}_2 \otimes \mathbf{D} \end{bmatrix}. \quad (80)$$

To facilitate computation, the denominator of (54) can be rewritten as

$$\begin{aligned} &\det \{ \mathbf{I}_L - 2\xi \mathbf{A} \mathbf{W} \} \\ &\times \det \{ \mathbf{I}_L + 2\xi \mathbf{D} \mathbf{W} + (2\xi)^2 \mathbf{F}^T \mathbf{W} [\mathbf{I} - 2\xi \mathbf{A} \mathbf{W}]^{-1} \mathbf{F} \mathbf{W} \} \\ &= \det \{ \mathbf{I}_L - 2\gamma_{\text{snr}} \xi \mathbf{R}(0) \boldsymbol{\Psi} \} \det \{ \mathbf{I}_L + 2\xi \mathbf{D} \mathbf{W} \\ &\quad + (2\xi)^2 \gamma_{\text{snr}} \mathbf{R}(z)^T \boldsymbol{\Psi} [\mathbf{I} - 2\gamma_{\text{snr}} \xi \mathbf{R}(0) \boldsymbol{\Psi}]^{-1} \mathbf{F} \mathbf{W} \} \end{aligned} \quad (81)$$

where $\mathbf{W} = 2\gamma_{\text{snr}}^2 [\mathbf{I}_L + 2\gamma_{\text{snr}} \boldsymbol{\Psi} \mathbf{R}(0)]^{-1} \boldsymbol{\Psi}$ and $\mathbf{R}(z)$ is defined in Sec. IV-A.

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